Towards Autonomous Air Cargo Network Design: A Measured Matheuristic over Branch-and-Price at Integrator Scale

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Code, instance generators and experiment logs:
`https://github.com/csotogd/flight_scheduling`

Abstract

Integrated airline schedule design — which flights to operate, which fleet flies each, how aircraft rotate, how cargo routes — remains planner-in-the-loop: state-of-the-art exact methods optimize over a candidate flight list that a human expert must write, and stall entirely at the scale of a global express integrator. Both facts are bottlenecks to automation. We present an approach that addresses them together, and a working system that measures it. Building on the branch-and-price-and-cut of Derigs and Friederichs (OR Spectrum 35:325–362, 2013), an autonomous design loop generates its own candidate flights round by round — propose, re-optimize, keep what pays, evict what does not — while a matheuristic layer keeps the optimization moving at scale: a *deliver-all* service model that prices contracted delivery as recourse and makes every subproblem feasible by construction; greedy cover seeding monetized by one warm LP over freshly priced columns; an escalating local-branching ball for integer adoption; and a geographic fix-and-optimize decomposition whose cross-boundary shipments carry *exact* connection windows read from the frozen timetable, so regional improvements splice into a globally verified schedule and the cycle is monotone by construction. On a synthetic integrator-scale instance (29,819 shipments, 1,250 flights, 154 aircraft), the whole-network integer search adopts nothing in ninety minutes while the regional cycle gains 4–5 M of weekly profit (synthetic units) in minutes; every component is toggleable and measured by ablation, including three negative results reported with equal prominence. Results are single-seed and exploratory; we state this, and the road to a full battery, explicitly.

1 Introduction

An express integrator sells a promise — everything tendered tonight is delivered across the planet within days — and keeps it with one of the most tightly coupled logistics systems in existence: hub-and-spoke networks whose feeders, overnight hub sorts, intercontinental trunks and morning distribution waves must interlock to the minute, flown by a heterogeneous fleet whose every aircraft chains through the week in a closed rotation. Planning such a network is traditionally cut into layers — which flights to operate, which fleet flies each, how aircraft rotate, how shipments route — optimized separately and reconciled by hand. The layers, however, interact strongly: a flight is worth operating only if cargo can reach it, cargo can reach it only if rotations position an aircraft, and an aircraft is free only if the rest of the

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schedule releases it. Sequential planning leaves structural money on the table, which is the classic argument for integrated optimization [15, 16].

Derigs and Friederichs [1] made that argument concrete for air cargo: they formulated the integrated problem (ACSP) over a cyclic week, gave a path/string column formulation, and solved it exactly by branch-and-price-and-cut (B&P&C) with problem-specific branching and implied-bound cuts, on four synthetic airline archetypes with up to a few thousand shipments.

Yet even the integrated method keeps a human in its loop, in two distinct ways. The first is by design: under the "pragmatic paradigm" the model *chooses from* a candidate flight list that an expert planner must write — the creative half of network design stays manual, does not scale with the network, and cannot run unattended. The second is empirical, and we establish it twice, independently, in Section 8.1: at integrator scale — tens of thousands of distinct origin–destination shipments per week over a thousand-plus flights, an order of magnitude above the literature's instances — the *integer* layer of the exact method stalls completely. From a good incumbent, ninety further minutes of whole-network branch-and-price explore one to three nodes and adopt nothing; a planner waiting for the tool gets silence. Figure 1 shows the anatomy behind the scale: most of an integrator's shipments are tiny and scattered, which drives much of what follows.

This paper brings an approach that attacks both bottlenecks together, so that network design can run *autonomously*: a design loop that generates, tests and retires its own candidate flights — no planner-written list — wrapped around a matheuristic layer that keeps the underlying optimization adopting improvements at a scale where the exact method alone has gone quiet. The planner's role moves from feeding the tool to reviewing its output.

Our answers are empirical. We reimplemented the full method of [1] (Section 2), generated instances from reimplementations of the paper's four archetype recipes plus a new public integrator-scale family, and then measured our way through a sequence of additions, keeping what worked and reporting what did not. The result is a matheuristic in the modern sense [13, 14]: heuristic scaffolding — greedy construction, large-neighborhood integer moves, spatial fix-and-optimize — around an exact column-generation engine that supplies the three things heuristics cannot: feasible *columns* for a timed, capacitated network; *dual prices* that tell every heuristic where value lies; and *valid bounds* that certify how much is left on the table.

Related work: the application. Air cargo operations have a rich literature — see the reviews of Feng et al. [16] on operations and Brandt and Nickel [19] on load planning — but it has traditionally treated the planning layers separately, in the tradition of service network design [15]: flight scheduling, fleet assignment, routing, each with its own models. Integrated air cargo optimization has become notably more active since: Zheng et al. [17] plan and schedule cargo networks under minimum-stay-time constraints with a matrix-based ALNS metaheuristic; Huang et al. [18] integrate aircraft and cargo *recovery* through a machine-learning-guided column-and-row generation on real carrier data; and joint fleet-scheduling and cargo-flow-allocation models solved by column-generation heuristics have appeared in the same period. The formulation of [1] remains the reference point for joint optimization of all four planning layers over a *cyclic week*.

What we have not found in that literature is either of the two things this paper targets. First, every one of these approaches optimizes over a flight set the planner supplies; none *generates and retires its own candidate flights*. Second, the reported computational studies stay one order of magnitude below the shipment counts of a global integrator, and none reports what happens to the integer layer there. Those are the two gaps this paper addresses — and we state them as gaps in *our* reading of the literature, not as a claim of exhaustive search.

Related work: the instruments. Methodologically we compose established components, each adapted to this problem's structure. Column generation and branch-and-price [2, 3, 4, 20]

supply priced itineraries, dual signals and valid bounds; under truncation we bound with Farley's construction [9], and against early dual oscillation we implement Wentges smoothing [7] in the stabilized-colgen tradition of du Merle et al. [8] — made mispricing-safe so certificates never degrade. For integer progress on masters too large to re-solve, we use local branching [5], a cousin of relaxation-induced neighborhoods [6], escalating the ball radius when flat. Our geographic decomposition belongs to the large-neighborhood and fix-and-optimize families [10, 11, 12], transplanted from variable space to *geography*; its novelty is what the cyclic weekly schedule forces: exact gateway windows read from the frozen timetable, balance-repaired partitions so both sides still assemble into rotations, and a monotone, independently verified merge. Our contribution is accordingly not a new component but a measured *composition*: which instruments still work when the master holds thirty thousand commodities, how they must be adapted, and — reported with equal prominence — which plausible instruments do *not* pay.

Contributions.

1. An *autonomous design loop* (Section 5) that removes the planner-written candidate list: four generator classes aimed at unserved demand, per-round solve–score–evict with amnesty, and the scale devices (demand consolidation, zone rotation) that keep rounds tractable — so the integrated model designs the network unattended and the planner reviews the result.

2. A *deliver-all* service model for integrators (Section 3): demand rows become equalities and every shipment carries an always-available contracted-delivery column priced at a multiple of own flying economics, derived from the mandatory schedule only so the tariff is identical across all model variants. Every relaxation, sub-MIP and decomposition block becomes feasible by construction, and the optimization becomes "own network versus contracting bill".

3. An integer-adoption pipeline for huge masters (Section 4): greedy cover seeding, flow monetization by a single warm LP over the freshly generated column pool, and an escalating local-branching ball [5] as the primal heuristic, with a per-round observable (*LP promise vs. integer adoption*) that separates "the candidate generator is weak" from "integer adoption is the bottleneck".

4. A geographic fix-and-optimize decomposition (Section 6) whose distinguishing features are *exact* gateway windows read from the frozen timetable (no estimated connection times), a balance-repair partition that keeps both the frozen and the re-optimized side assemblable into rotations, an exact fleet slice with a learned correction for chain-splitting rounding, and a monotone merge guard: a block result is accepted only when the independently verified *global* schedule improves. An optional pairwise "relay" re-opens cross-region demand with no cross-pass bookkeeping.

5. A measured engineering layer (Section 7): a bit-identical parallel pricing sweep, Farley-type bounds [9] under truncation with a membership filter, and per-iteration cost breakdowns that make the next bottleneck visible instead of guessed. Mispricing-safe dual smoothing [7, 8] is implemented and correctness-verified; its performance evaluation belongs to the planned battery and no benefit is claimed here.

6. An honest computational study (Section 8) on public, deterministic, regenerable instances, including three negative results: coordinated "hub wave" candidate bundles that the optimizer never adopts; cross-region relay passes that run clean but earn nothing at the chosen recourse tariff; and a specialized LP method (sifting) that loses to warm-started dual simplex on the colgen master.

We are explicit about scope (Section 9): headline numbers are single-seed, the instances

airports / transfer hubs	150 / 9
fleets (aircraft)	6 (154)
flights in base schedule (mandatory / optional)	1,250 (625 / 625)
shipments (O&D demands)	29,819
total demand	∼62.7 kt/week
service commitment	deliver-all (Section 3)

Table 1: The integrator-scale instance RLA-I-s1 (synthetic, seed 1).

are synthetic with AI-estimated parameters, and comparability with [1] is qualitative because neither their instances nor their code were published. This preprint plants the system, the mechanisms and the measurement discipline; the multi-seed battery and an arc-flow baseline are the declared next step. The remainder of the paper follows the pipeline: model and base algorithm (Sections 2–3), integer adoption and network design (Sections 4–5), geographic decomposition (Section 6), engineering (Section 7), results and limitations (Sections 8–9).

2 Problem and base algorithm

We summarize [1] only to fix vocabulary. A weekly cyclic horizon of $N = 10{,}080$ minutes carries: *airports* (some transfer hubs with minimum connection times, transfer and storage costs); *fleets* with counts, payload/volume capacities and costs; *flights* (sequences of legs; mandatory or optional; possibly external capacity with booking costs); and *shipments* (O&D demands with release times, delivery deadlines, tonnage, volume, freight rate).

Model ACSP-T is a path/string formulation over a time-space network: columns are cargo *paths* (feasible timed itineraries for one O&D) and flight *strings* (fleet-assigned flight sequences; in the FARP-T variant used throughout this paper a string is a single flight and rotations are chained through ground arcs). Rows enforce demand limits, leg payload and volume capacities coupled to string selection, flight cover, per-fleet flow balance at timeline events, and fleet counts. Columns are priced by two oracles: PRICE-P, a resource-constrained shortest path solved with A*; and PRICE-S for strings. Implied-bound cuts link flow to selection; problem-specific branching (flight in/out, fleet-flight, follow-on) keeps both pricers intact. Without maintenance the procedure is exact. Our reimplementation follows the paper; every accepted incumbent is re-validated by an independent feasibility checker that shares no code with the solver.

Instances. We reimplemented the generation recipes of the four archetypes of [1] (regional, international, mixed, express) from the published parameter tables, and added two fictional families: GI ("global integrator": 4 hubs, 140 aircraft, dense demand) and RLA ("real-life-scale airline": 9 hubs on three continents, 154 aircraft in six fleets, gravity-model demand on 60% of ordered station pairs with weekly lane recurrence, 1,250-flight base schedule of which half is mandatory, and 29,819 shipments). All parameters are AI-estimated for plausibility; no real airline data is used. Generators are deterministic in an integer seed and public. Table 1 summarizes the instance RLA-I-s1 used for the scale experiments.

3 A deliver-all service model

Integrators do not cherry-pick profitable freight; a service commitment means everything tendered is delivered, if necessary by a contracted external carrier. We model this by (i) turning demand rows into equalities and (ii) giving every shipment od a *contracted delivery column* with objective coefficient $r_{od} - c^{\text{rec}}_{od}$, upper bound d_{od}, and a single nonzero entry in the demand

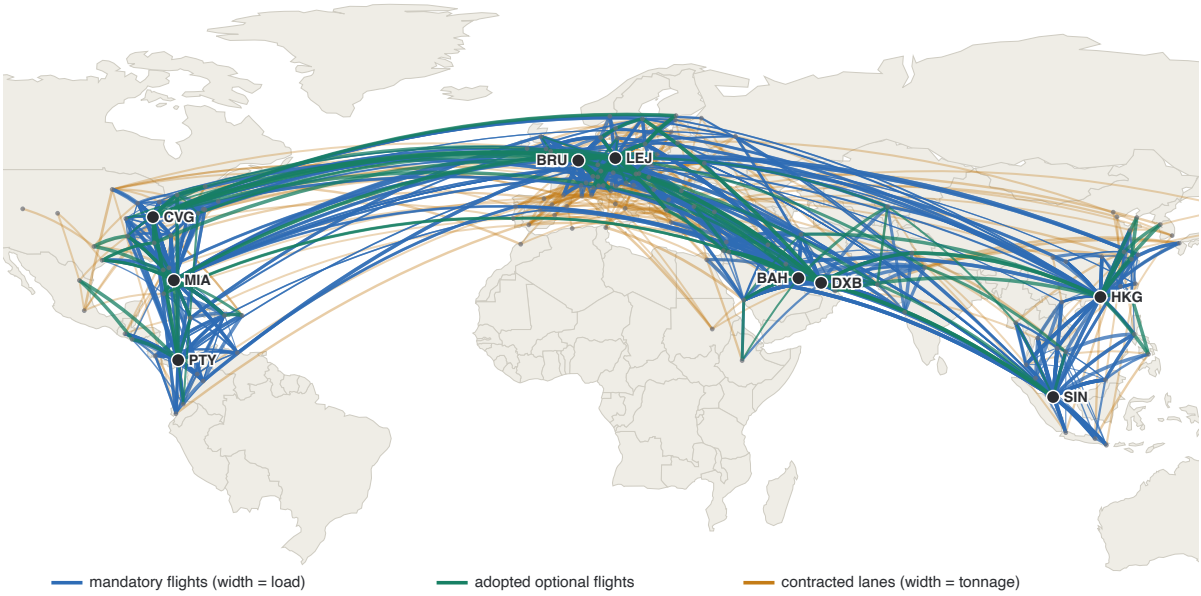

Figure 1: The best full-demand schedule on RLA-I-s1, drawn to scale. Blue: mandatory flights (width = load); green: adopted optional flights; amber, underneath: the 250 heaviest *contracted* lanes (width = tonnage) — demand served by paid external carriers. The amber fans around the hubs are the 46.6 M contracting bill of Section 8.2 made visible, and the target of the wave, relay and decomposition experiments.

row. The recourse tariff is

$$c^{\text{rec}}_{od} \;=\; \mu \cdot \big(\delta_{od}\, v + \delta_{od}\, \phi / W\big),$$

with δ_{od} the great-circle distance, v the average own variable cost per tonne-km, ϕ the median own fixed cost per km, W the largest fleet payload, and $\mu = 3$ throughout. Crucially, v and ϕ are derived from the *mandatory* flights only: that set is identical in every model variant of a design or decomposition run (optional candidates come and go), so the tariff — and hence the profit scale — is stable across all comparisons. An earlier version derived the tariff from all flights and suffered a $\sim$1% per-round drift that silently invalidated round-over-round comparisons.

Three consequences matter. Revenue becomes a constant, so the problem is a cost minimization (own flying vs. contracting bill). Every selection of flights is completable, so seeds, restricted sub-MIPs and decomposition blocks are feasible by construction — the entire pipeline of Sections 4 and 6 leans on this. And the recourse coefficient caps the demand duals at the tariff, a free mild stabilization of the master.

4 Integer adoption at scale

At integrator scale the practical failure is primal: branch-and-bound explores 1–3 nodes within realistic budgets (Section 8.1), so incumbents must come from elsewhere. Our pipeline has four stages, each measured in isolation in Table 2.

(a) Greedy cover seed. A constructive pass assigns every mandatory flight to its lightest compatible fleet as balanced round trips (inter-hub mirror pairs kept together), followed by an evaluated repair loop. The seed is feasible but flowless: it earns no revenue (deliver-all: everything is contracted).

(b) Seed flow loader. After the root column generation, the seed's selection is fixed in the master and *one* warm LP optimizes cargo flows over the column pool the pricers just generated. This is where column generation pays its first dividend: the pool contains thousands of priced itineraries, and one LP monetizes the skeleton in seconds. No new pricing is needed.

(c) Escalating local-branching ball. The primal heuristic solves the restricted master MIP inside a Hamming ball of at most k selection flips around the incumbent [5], escalating $k \to 2k \to 4k$ (default $k = 60$) while flat within a budget slice. The incumbent is always feasible inside the ball, so the step is monotone; the solver, not a hand-built neighborhood, chooses *which* flips, which keeps cross-network rotation rewirings reachable. An unrestricted restricted-master MIP at this scale finds nothing in equal budget (Table 2); the ball turns the same budget into adopted profit. On adoption, flows are re-monetized by one further warm LP.

(d) Promise vs. adoption. Each design round logs the LP bound uplift contributed by the current candidate batch (*promise*) against the integer profit actually adopted. This single observable separated two hypotheses we could not otherwise distinguish — "the proposer generates worthless candidates" vs. "the integer layer cannot digest them" — and later diagnosed a genuine economic no (Section 8.4).

5 The autonomous design loop

The base algorithm assumes a human writes the optional-flight candidate list. Our design loop generates it: each round proposes a batch of candidate flights aimed at unserved demand (hub round trips, direct rotations, inter-hub trunks, premium external capacity), solves the extended instance with warm-started columns and a seeded incumbent, scores which candidates flew, and evicts persistent non-flyers (with amnesty). Two scale devices are toggleable: consolidation of tiny, far shipments into hub-corridor pseudo demands during the rounds (the final solve always runs on the original demand), and zone rotation of proposal targeting. We omit details for space; all of it is in the repository's technical report. One addition is a negative result we keep in the codebase, disabled: *hub waves*, coordinated feeder–trunk–distribution bundles proposed as a set for the long tail of scattered small shipments. The optimizer never adopted one (Section 8.4).

6 Geographic fix-and-optimize with exact gateway windows

Motivated by the observation that block-sized models adopt improvements in seconds while the whole-network master adopts nothing (Section 8.3), we decompose the planet: hub clusters (hubs within 3,000 km) define regions; every airport joins its nearest hub's region.

Blocks. A block re-optimizes one region against a frozen world. Re-decidable (*kept*) flights are those with every leg inside the region, minus three repair classes computed to a fixpoint: (i) *balance repair* — the kept *selected* strings must balance arrivals and departures per (fleet, airport), or neither side of the partition assembles into rotations; unbalanced one-way strings (typically a multi-stop trunk whose mirror has an en-route stop in a third region) are frozen one at a time, so the cascade follows the unbalanced chain and stops; (ii) flows whose path crosses the region more than once, and (iii) degenerate cuts (a loop entering and leaving at the same airport), whose flights are frozen because the merge splices exactly one regional segment per donated flow.

Exact windows. Cross-boundary cargo is truncated at its gateway with windows read from the *frozen timetable*, not estimated: an outbound segment must end before its frozen onward connection departs minus the gateway transfer time; an inbound segment becomes available when its frozen feeder lands plus transfer. Cargo handling at true origins and destinations is encoded into the windows so gateway hand-offs do not pay it twice. Consequently a time-feasible block solution *splices* back into a time-feasible global schedule by construction. Shipments with both endpoints inside the region enter whole; under deliver-all their contracted remainder rides along, so every block is feasible from its seed.

Fleet slice. The block's fleet budget is count$-$used(all)$+$used(kept), computed by assembling the kept side alone (possible after balance repair). Splitting mixed rotation chains can cost a rounding aircraft per fragment that no a-priori budget foresees; instead of estimating it, the cycle *learns* it: a merged block that fails the global fleet-size check by e_k aircraft teaches its block a budget reduction of e_k for the next visit.

Monotone merge. A block solution is spliced into the global schedule (frozen strings and flows kept; donated segments replaced; sub-contracted remainders fall back to global end-to-end contracting) and accepted *only if* the independently verified global profit improves. The cycle can therefore never lose ground. During development this guard caught four distinct merge bugs — double-counted multi-crossing segments, duplicated whole-region demands, unassemblable partitions, fleet-rounding violations — without ever letting a corrupted schedule through; we consider it the load-bearing design decision of the whole section.

Pairwise relay. Optionally, after the single-region passes of each cycle, *pair* passes run on region unions ordered by open cross-tonnage. A cross-region shipment enters a pair block whole, with exact windows; feeder, trunk and distribution are decided in one model, avoiding any cross-pass ledger or estimated hand-off times.

7 Engineering for honest measurement

Parallel pricing, bit-identical. Each pricing iteration asks one independent shortest-path question per shipment against the same duals. The sweep runs one shipment per core and collects results in shipment order, so its output is bit-identical to the sequential sweep (asserted by test and verified in production runs: identical column counts and bounds). The per-destination A* bounds — the only lazily built shared state — are prewarmed one Dijkstra pair per core. The servability probes of the design loop parallelize the same way with thread-local dual vectors.

Mispricing-safe dual smoothing. Optionally, pricers see $\alpha \cdot \text{center} + (1-\alpha) \cdot \text{duals}$ [7, 8] ($\alpha = 0.7$). Two safeguards keep exactness: a smoothed pass that finds no columns triggers a re-price with the true duals before any conclusion, and validity-critical bounds are computed from true-dual passes only. A test asserts that the stabilized root converges to the same LP value as the plain one. We report no performance numbers for smoothing: it is implemented, verified and off by default, and its evaluation is deliberately left to the multi-seed battery — we flag this to avoid claiming a benefit we have not measured.

Bounds under truncation. Hard per-round budgets are made mathematically legal by a Farley-type bound [9]: on interruption the tightest valid bound seen across full pricing passes is returned, each computed as the RMP value plus the most optimistic contribution of *missing* columns. A membership filter is essential in practice: pricers legitimately return columns

stage	profit (M)	increment
greedy cover seed (all demand contracted)	129.9	—
+ seed flow loader (one warm LP over the pool)	139.6	+9.7
+ local-branching ball k=60	142.3	+2.7
+ adoption re-monetization (one warm LP)	143.3 / 143.5	+1.0
unrestricted restricted-master MIP, same budget	139.6	+0.0

Table 2: Round-0 incumbent waterfall on RLA-I-s1 (consolidated model, deliver-all). The re-monetization column shows two independent runs. The last row is the ablation: without the ball, the same heuristic budget adopts nothing.

already in the master (nonbasic at bounds, positive reported reduced cost); counting them produces absurd bounds. Only columns not yet in the master contribute.

Cost visibility. Every colgen iteration reports its breakdown (pricing / column loading / LP re-solve seconds). This 20-line addition settled in minutes a question we had been guessing at for days (Section 8.5).

8 Computational results

Protocol and disclosure. All numbers below are from single runs on one machine (Apple Silicon, 10 cores; CPLEX 22.2 as LP/MIP backend, HiGHS as fallback), on instance RLA-I-s1 unless stated. They are exploratory: no variance estimates, one seed. We report them because their margins are large and their patterns repeated across the campaign; the multi-seed battery over all instance families is the declared next step. All runs are reproducible from the public repository (deterministic seeds; every figure's raw log is committed).

8.1 The whole-network integer layer stalls

From a shared incumbent at 139.7–139.8 M (produced by cover seed + loader), the whole-network solver — warm-started column pool, seeded incumbent, escalating ball, 90-minute budget — produced *zero* incumbent improvements, twice (independent runs at 1,800 s and 5,400 s produced the same +0). Branch-and-bound explored 1–3 nodes. This is the phenomenon the rest of the paper responds to: at this scale the exact tree is vestigial and even ball-restricted sub-MIPs drown in the full master's LP.

8.2 The adoption pipeline

Table 2 and Figure 2 show the round-0 waterfall on the consolidated design model (25,451 demands after tiny-far consolidation). Each stage is enabled on top of the previous.

The best full-demand schedule obtained by the design pipeline alone reaches 142.8 M at a 9.9% certified gap; in it, 22,011 of 29,819 shipments (46.0 kt, a 46.6 M bill) are contracted and only 58 of 154 aircraft fly. That anatomy — a long tail of ∼2-tonne shipments scattered over ∼150 stations, the amber fans of Figure 1 — motivates both the wave experiment (negative, below) and the decomposition (positive, next).

8.3 Geographic decomposition

A/B protocol: one shared base incumbent, then the same additional budget spent either (A) continuing the whole-network solve or (B) running the regional cycle. Table 3 summarizes four consecutive runs (two clean A/Bs plus two earlier runs whose arm B was invalidated by merge

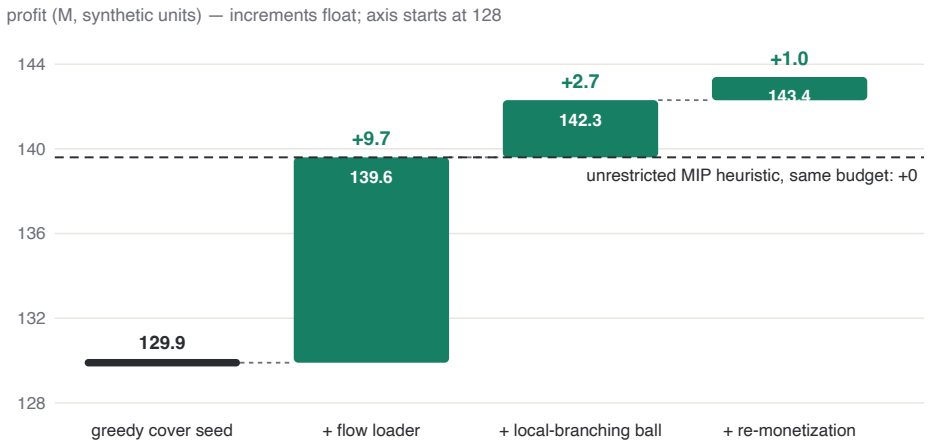

Figure 2: The adoption pipeline of Section 4 as a waterfall (data of Table 2). Green segments are the increments each stage adds on top of the previous; the dashed line is the ablation — the unrestricted restricted-master MIP, given the same budget as the ball, never leaves the loader's level.

arm	budget	gain	note
A: whole-network continuation	1,800 s	+0	
A: whole-network continuation	5,400 s	+0	independent repeat
B: regional cycle	900 s	+5.03 M	6 accepted blocks, then flat
B: regional cycle (learned fleet shave)	1,800 s	+3.96 M	includes learning visits

Table 3: Whole-network vs. regional split from a shared incumbent (RLA-I-s1, full 29,819-shipment demand). Blocks of 200–350 flights adopt improvements in 4–50 seconds each.

bugs — in *every* run, including the buggy ones, the global feasibility guard held: no corrupted schedule was ever accepted).

The mechanism, not just the outcome, matches the diagnosis: all four regions adopt on their first visit (from +0.26 M to +1.5 M each, in 9–50 s of block time), improvements shrink on the second cycle, and the cycle then honestly reports flat blocks until the budget closes. The best full-demand schedule of the campaign (144.8 M) came from a 5-minute base solve plus a 15-minute regional cycle.

8.4 Negative results

Hub waves. A fifth candidate class proposed coordinated feeder–trunk–distribution bundles per (hub corridor, day), with chain-consistent times and honestly windowed tonnage. Over 20 design rounds (6,000 candidates, waves included at 30% quota), *zero* were adopted, and the LP promise of candidate batches decayed from +4.4 M (round 1) to ∼0 (round 3 on). *Pair relay.* The pairwise passes of Section 6 ran clean (no crashes, all merges verified) and adopted nothing at $\mu = 3$. *Interpretation.* Three independent probes — waves, promise decay, pairs — agree: at triple own economics, the scattered cross-region tail is genuinely cheaper to contract than to fly. This is a statement about the instance's economics, not about the algorithms; both mechanisms remain in the code, disabled by default, for cheaper-recourse or larger-scale settings.

LP method. On the colgen master (> 40k columns), CPLEX sifting — nominally suited to column-heavy LPs — measured 2–3× *slower* per re-solve (49–114 s) than warm-started dual simplex (18–44 s). The advantage of basis reuse across colgen iterations is well known; the value

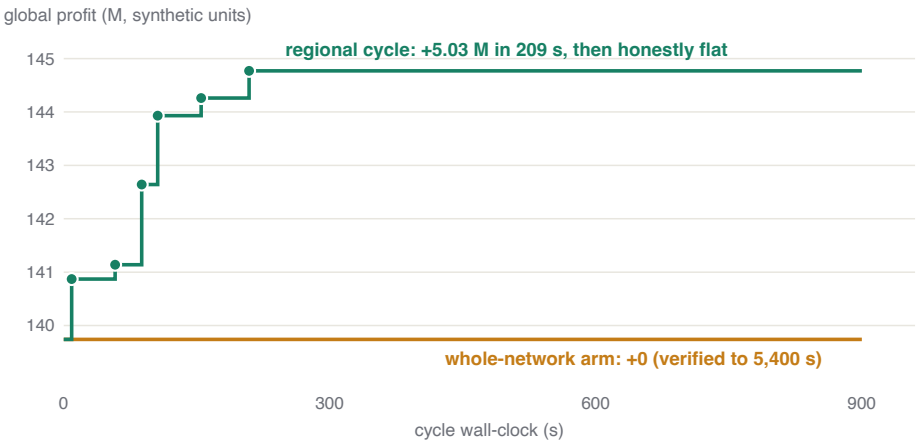

Figure 3: The clean A/B of Table 3, over the cycle clock. Each dot is an accepted block merge (globally verified); the plateau after 209 s is the cycle reporting flat blocks rather than manufacturing noise. The whole-network arm, from the same incumbent, adopts nothing in 5,400 s.

here is quantifying it for this master, since it rules out a drop-in remedy for the LP bottleneck of Section 8.5.

8.5 Parallel pricing and the next bottleneck

Parallelizing the pricing sweep cut it from ∼25 s to 0.4–1.0 s per iteration (identical columns and bounds), but end-to-end root time improved only ∼15% (156 s → 133 s for six iterations). The per-iteration breakdown explains it: pricing 1.0 s, column loading 0.1 s, master LP re-solve 18–44 s and growing with the column count. Amdahl's ledger is now unambiguous: the master LP is the next bottleneck, and the natural lever is column deletion (keeping the master lean) rather than a faster pricer. Dual smoothing also gains value from this shift: every avoided iteration now saves a full LP re-solve.

9 Limitations

(i) Headline results are single-seed, single-machine, one instance family at scale; margins are large but variances are unmeasured. (ii) All instances are synthetic; parameters (costs, demands, fleets, curfews, payload-range curves) are AI-estimated for plausibility and calibrated in scale — not in detail — to [1]; no real airline data is used or implied. (iii) Comparability with [1] is qualitative only: their instances and code were not published, so our reimplementation of their generators matches parameters, not files. (iv) There is no external baseline yet (e.g., a direct arc-flow MIP under equal budget); "better than our own previous configuration" is the only comparison this preprint supports. (v) The design loop is greedy across rounds; inner solves carry valid bounds, but no global optimality claim spans rounds.

10 Conclusions and road map

At integrator scale, the exact tree of B&P&C contributes almost nothing, but the column-generation engine remains the value chain's spine: every profitable move in our campaign flowed through priced columns, warm LPs over them, or dual signals derived from them. The productive question is not "exact or heuristic" but *where each layer earns its keep*: LP/colgen for columns, prices and certificates; local branching for integer moves the full master cannot

digest; geographic fix-and-optimize for turning block-sized tractability into global, verified, monotone progress.

Road map, in order: (1) the multi-seed battery over all instance families with component ablations; (2) an arc-flow MIP baseline under equal budget; (3) column deletion for the master LP; (4) deeper trees inside regional blocks, where nodes cost seconds; (5) any form of real-data validation, which would upgrade every claim in this paper.

Reproducibility and AI disclosure

The complete implementation (solver, generators, web workbench, 120 verification tests), all instance generators, and the raw logs behind every number in this paper are public in the accompanying repository. Runs are deterministic given a seed. This project was developed with substantial assistance from an AI coding agent (Anthropic Claude), including code, instance parameter estimation, experiment execution and the drafting of this manuscript; all algorithmic decisions, measurements and claims were verified by the reproducible pipeline above, and the human author reviewed and takes responsibility for the content. Airline names and data are fictional.

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